

Topic Unit 1 - AP Challenge Review

AP Challenge Review

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

This cumulative review integrates Topics 1.1-1.14. Questions are original AP-style classroom tasks. Use a calculator only when the item permits it. Show complete reasoning on free-response questions and preserve domain restrictions.

Answer grid

1 _____	2 _____	3 _____	4 _____	5 _____	6 _____	7 _____	8 _____
9 _____	10 _____	11 _____	12 _____	13 _____	14 _____	15 _____	16 _____

Multiple Choice - Part 1

MCQ 1

No calculator

Which relation fails to define y as a function of x ?

- A. $(1, 4), (2, 4), (3, 7)$
- B. $(-1, 2), (0, 3), (-1, 5)$
- C. $(0, 0), (1, 1), (2, 4)$
- D. $(-2, 5), (2, 5), (4, 1)$

MCQ 2

No calculator

For $f(x) = x^2 - 3x + 2$, the average rate of change on $[1, 5]$ is

- A. 1
- B. 3
- C. 4
- D. 6

MCQ 3

No calculator

At equally spaced inputs, outputs are 4, 9, 18, 31, 48. Which evidence is strongest?

- A. Constant first differences
- B. Constant second differences
- C. Constant ratios
- D. Constant residuals

MCQ 4

No calculator

Which polynomial has left end up and right end down?

- A. $2x^6 - x$
- B. $-3x^6 + x^2$
- C. $-x^5 + 4x$
- D. $x^5 - 7$

Multiple Choice - Part 2

MCQ 5

No calculator

A real-coefficient polynomial has zero $2 - 5i$. Which zero is required?

- A. $-2 + 5i$
- B. $2 + 5i$
- C. $-2 - 5i$
- D. $5 - 2i$

MCQ 6

No calculator

The end behavior of $R(x) = (3x^4 - x)/(2x^4 + 7)$ is described by

- A. $y = 0$
- B. $y = 3/2$
- C. $y = 2/3$
- D. no horizontal asymptote

MCQ 7

No calculator

For $R(x) = (x - 2)(x + 1)/[(x - 2)(x - 5)]$, which statement is true?

- A. $x = 2$ is a zero
- B. $x = 2$ is a vertical asymptote
- C. $x = 2$ is a hole
- D. $x = 5$ is a hole

MCQ 8

No calculator

Near $x = 3$, $R(x) = -4/(x - 3)^2$ approaches

- A. $+\infty$ from both sides
- B. $-\infty$ from both sides
- C. $-\infty$ left and $+\infty$ right
- D. $+\infty$ left and $-\infty$ right

Multiple Choice - Part 3

MCQ 9

No calculator

The graph of $(x^2 - 4)/(x - 2)$ has a hole at

- A. (2, 0)
- B. (2, 2)
- C. (2, 4)
- D. (-2, 0)

MCQ 10

No calculator

The remainder when $P(x) = x^3 - 4x + 7$ is divided by $x + 2$ equals

- A. -9
- B. -1
- C. 7
- D. 23

MCQ 11

No calculator

If $(6, -2)$ lies on f , which point lies on $g(x) = -3f(2(x - 1)) + 5$?

- A. (2, 11)
- B. (4, 11)
- C. (4, 1)
- D. (13, 11)

MCQ 12

No calculator

An observed value is 51 and the model predicts 56. The residual is

- A. -5, an overprediction
- B. -5, an underprediction
- C. 5, an overprediction
- D. 5, an underprediction

Multiple Choice - Part 4

MCQ 13

No calculator

Which data pattern most strongly supports exponential growth?

- A. Outputs add 8 each step
- B. First differences add 3 each step
- C. Outputs multiply by 1.2 each step
- D. Residuals are all positive

MCQ 14

No calculator

A quadratic has vertex $(3, 5)$ and passes through $(5, -3)$. Its leading scale is

- A. -2
- B. $-1/2$
- C. $1/2$
- D. 2

MCQ 15

No calculator

A model counts assembled devices during one shift, with capacity 420. Which is the most defensible domain?

- A. all real numbers
- B. $0 \leq x \leq 420$ for real x
- C. integers $0 \leq x \leq 420$
- D. $x > 420$

MCQ 16

No calculator

Which statement correctly compares $f(x) = (x^2 - 1)/(x - 1)$ and $g(x) = x + 1$?

- A. They are equal for every real input
- B. They agree only at $x = 1$
- C. They agree for $x \neq 1$, but their usual domains differ
- D. They have different values for all inputs

AP-style free response 1

Calculator permitted - 6 points

A production index is recorded at equally spaced inputs.

x	0	1	2	3	4	5
$P(x)$	5	10	19	32	49	70

- A. Use numerical structure or regression to construct a model for $P(x)$.

- B. Find and interpret the average rate of change on $[1, 4]$.

- C. Solve $P(x) = 100$ and report the relevant input to three decimals.

- D. At $x = 6$, an observed value is 94. Compute and interpret its residual, then state one limitation of predicting beyond the table.

AP-style free response 2

No calculator - 6 points

Consider

$$R(x) = \frac{(x-3)^2(x+1)(x-2)}{(x-2)(x+4)^2}.$$

A. Simplify R and state every original domain restriction.

B. Identify all zeros, holes, and vertical asymptotes, including multiplicity behavior at each zero.

C. Determine both one-sided limits at the vertical asymptote.

D. Use polynomial division to find the slant asymptote and explain why it describes end behavior.
